

Experiment – 16

High Level Programming Extensions – Procedures

```
mysql> USE ProcedureDB;
Database changed
mysql> CREATE TABLE Student (
  ->     StudentID INT PRIMARY KEY,
  ->     StudentName VARCHAR(50),
  ->     Department VARCHAR(50),
  ->     Marks INT
  -> );
Query OK, 0 rows affected (0.11 sec)

mysql> INSERT INTO Student VALUES
  -> (101, 'Indhu', 'CSE', 85),
  -> (102, 'Meghana', 'IT', 78),
  -> (103, 'Shalini', 'ECE', 92),
  -> (104, 'Teju', 'CSE', 67);
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
|      101 | Indhu       | CSE       | 85    |
|      102 | Meghana     | IT        | 78    |
|      103 | Shalini     | ECE       | 92    |
|      104 | Teju        | CSE       | 67    |
+-----+-----+-----+-----+
4 rows in set (0.04 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetStudents()
  -> BEGIN
  ->     SELECT * FROM Student;
  -> END //
Query OK, 0 rows affected (0.09 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> CALL GetStudents();
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
|      101 | Indhu       | CSE        | 85    |
|      102 | Meghana     | IT         | 78    |
|      103 | Shalini     | ECE        | 92    |
|      104 | Teju        | CSE        | 67    |
+-----+-----+-----+-----+
4 rows in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetStudentsByDepartment(IN dept VARCHAR(50))
-> BEGIN
->     SELECT * FROM Student
->     WHERE Department = dept;
-> END //
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetStudentsByDepartment('CSE');
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
|      101 | Indhu       | CSE        | 85    |
|      104 | Teju        | CSE        | 67    |
+-----+-----+-----+-----+
2 rows in set (0.05 sec)

Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CALL GetStudentsByDepartment('IT');
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
|      102 | Meghana     | IT         | 78    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetStudentsAboveMarks(IN m INT)
-> BEGIN
->     SELECT StudentID, StudentName, Marks
->     FROM Student
->     WHERE Marks > m;
-> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetStudentsAboveMarks(80);
+-----+-----+-----+
| StudentID | StudentName | Marks |
+-----+-----+-----+
|      101 | Indhu       | 85    |
|      103 | Shalini     | 92    |
+-----+-----+-----+
2 rows in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SHOW PROCEDURE STATUS WHERE Db = 'Procedure00';
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Db      | Name                | Type      | Definer      | Modified      | Created      | Security_type | Comment      | character_set_client | collation_connection | Database Collation |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| proceduredb | GetStudents          | PROCEDURE | root@localhost | 2026-08-11 11:19:01 | 2026-08-11 11:19:01 | DEFINER      |              | cp850                | cp850_general_ci    | utf8mb4_0900_ai_ci |
| proceduredb | GetStudentsAboveMarks | PROCEDURE | root@localhost | 2026-08-11 11:20:08 | 2026-08-11 11:20:08 | DEFINER      |              | cp850                | cp850_general_ci    | utf8mb4_0900_ai_ci |
| proceduredb | GetStudentsByDepartment | PROCEDURE | root@localhost | 2026-08-11 11:19:36 | 2026-08-11 11:19:36 | DEFINER      |              | cp850                | cp850_general_ci    | utf8mb4_0900_ai_ci |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.06 sec)

mysql> DROP PROCEDURE GetStudents;
Query OK, 0 rows affected (0.05 sec)

mysql> DROP PROCEDURE GetStudentsByDepartment;
Query OK, 0 rows affected (0.06 sec)
```